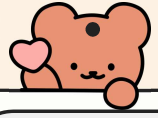
My Nervous PPT

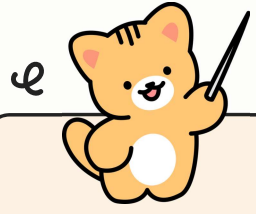


USB 패킷 포렌식 이해



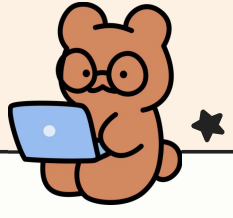
1. USB HID 패킷 분석
2. HID 키 입력 복원
3. 패킷 시간 간격을 활용한 데이터 해석

My Nervous PPT



문제 설명

Umm.. 난 PPT 발표를 하기 전, 긴장된 마음을 풀고자
프리젠터를 막 누르곤 하지.. :(



My Nervous PPT

풀이과정

```
INTERFACE DESCRIPTOR (0.0): class HID
    bLength: 9
    bDescriptorType: 0x04 (INTERFACE)
    bInterfaceNumber: 0
    bAlternateSetting: 0
    bNumEndpoints: 1
    bInterfaceClass: HID (0x03)
    bInterfaceSubClass: Boot Interface (0x01)
    bInterfaceProtocol: Keyboard (0x01)
    iInterface: 0
```

USB 트래픽에서 Device 32의 HID 입력 장치를 식별

```
ENDPOINT DESCRIPTOR
    bLength: 7
    bDescriptorType: 0x05 (ENDPOINT)
  > bEndpointAddress: 0x81  IN  Endpoint:1
  > bmAttributes: 0x03
  > wMaxPacketSize: 64
    bInterval: 10
```

프리젠터 입력이 전달되는 Endpoint 0x81을 찾음

```
USB URB
    [Source: 2.32.1]
    [Destination: host]
    USBPcap pseudoheader length: 27
    IRP ID: 0xffffae0e0f7567d0
    IRP USBD_STATUS: USBD_STATUS_SUCCESS (0x00000000)
    URB Function: URB_FUNCTION_BULK_OR_INTERRUPT_TRANSFER (0x0009)
  > IRP information: 0x01, Direction: PDO -> FDO
    URB bus id: 2
    Device address: 32
  > Endpoint: 0x81, Direction: IN
    URB transfer type: URB_INTERRUPT (0x01)
    Packet Data Length: 8
    [bInterfaceClass: HID (0x03)]
HID Data: 0200520000000000
```

```
USB URB
    [Source: 2.32.1]
    [Destination: host]
    USBPcap pseudoheader length: 27
    IRP ID: 0xffffae0e0f7567d0
    IRP USBD_STATUS: USBD_STATUS_SUCCESS (0x00000000)
    URB Function: URB_FUNCTION_BULK_OR_INTERRUPT_TRANSFER (0x0009)
  > IRP information: 0x01, Direction: PDO -> FDO
    URB bus id: 2
    Device address: 32
  > Endpoint: 0x81, Direction: IN
    URB transfer type: URB_INTERRUPT (0x01)
    Packet Data Length: 8
    [Request in: 32]
    [Time from request: 0.327954000 seconds]
    [bInterfaceClass: HID (0x03)]
HID Data: 0200510000000000
```

0x51, 0x52 값이 반복되는 것을 확인

My Nervous PPT

풀이과정

값	입력	변환
0x51	↓	.
0x52	↑	-

두 버튼 입력을 . 과 - 에 대응시켜 모스 부호로 변환
패킷 사이의 긴 시간 간격을 기준으로 입력을 문자 단위로 분리

```
-.- .... -.- -.--. .-.. --- --- -.- ..--.- .- - ..--.- --
-.-- ..--.- .--. .--. - -.--.-
```

디코딩 결과

모스 부호 KHK(LOOK

최종 Flag

KHK(LOOK

사용 도구

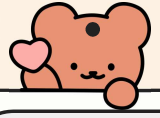


Wireshark

모스부호 변환 사이트

https://dencode.com/ko/string/morse-code

Take Your Glasses



다양한 스테가노그래피 툴 연습

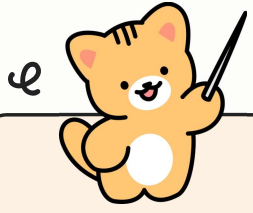


① PNG 파일 구조를 확인하는 방법

② 손상된 이미지의 크기 값을 복구하는 방법

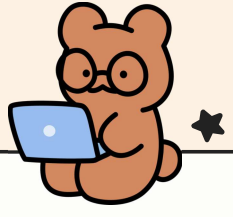
③ 복구된 이미지에서 숨겨진 문자열을 찾는 방법

Take Your Glasses



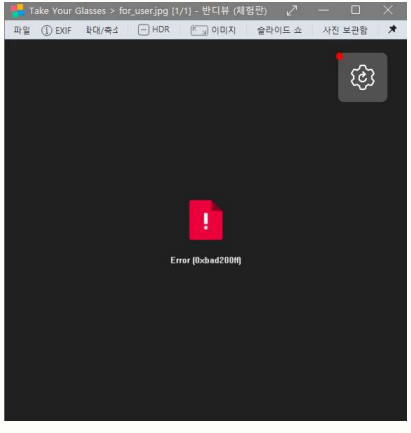
문제 설명

안경을 쓰고, 두 눈을 똑바로 뜨세요!

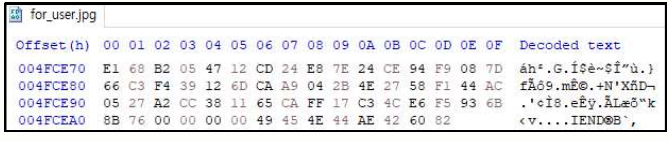


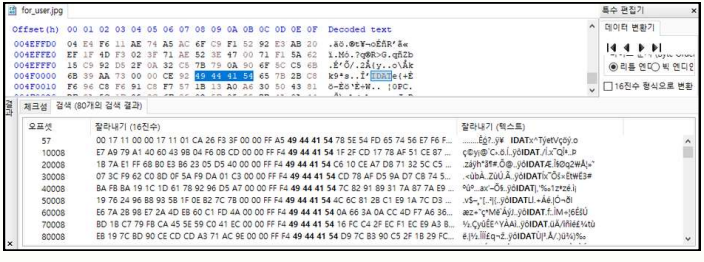
My Nervous PPT

풀이과정



주어진 JPG 이미지를 열어보면 오류가 발생함





파일 확장자는 JPG지만 내부 구조에서 PNG chunk 발견

→ 실제 파일 형식이 PNG임을 추정

My Nervous PPT

풀이과정

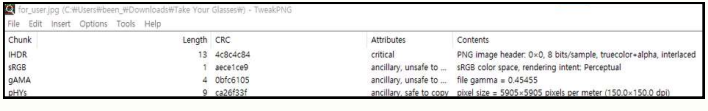
for_user.jpg

Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	89	50	4E	47	0D	0A	1A	0A	00	00	00	0D	49	48	44	52	‰PNG........IHDR
00000010	00	00	00	00	00	00	00	00	08	06	00	00	01	D4	80	35	..............Ô€5
00000020	FB	00	00	00	01	73	52	47	42	00	AE	CE	1C	E9	00	00	û....sRGB.®Î.é..

89 50 4E 47 0D 0A 1A 0A | 00 00 00 0D | 49 48 44 52

PNG 시그니처 IHDR 길이(13바이트) IHDR

PNG 파일의 앞 16바이트는 고정값이므로
해당 값으로 헤더 일부 복원



IHDR의 Width와 Height가 0×0으로 손상된 것을 확인

```
2026-08-25 14:43:41 DESKTOP-S7QLHGT inspect_png.chunks[8104] WARNING Calculated crc of b'IHDR' does not match
2026-08-25 14:43:41 DESKTOP-S7QLHGT inspect_png.fixing[8104] INFO Creating C brute forcer:
2026-08-25 14:43:41 DESKTOP-S7QLHGT inspect_png.fixing[8104] INFO Found dimensions: 1500 x 1125
IHDR (1):
[8-32]
Chunk: b'\x00\x00\x05\xdc\x00\x00\x04e\x08\x06\x00\x00\x01'
Width:             1500
Height:            1125
Bit depth:         8
Color type:        6
Compression method: 0
Filter method:     0
Interlace method:  1
```

png-tools를 이용해 IHDR CRC 기반 브루트포스로
이미지 크기 1500 × 1125 확인

for_user.png

Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	89	50	4E	47	0D	0A	1A	0A	00	00	00	0D	49	48	44	52	‰PNG........IHDR
00000010	00	00	05	DC	00	00	04	65	08	06	00	00	01	D4	80	35	...Ü...e.....Ô€5
00000020	FB	00	00	00	01	73	52	47	42	00	AE	CE	1C	E9	00	00	û....sRGB.®Î.é..

손상된 Width/Height 값 복구

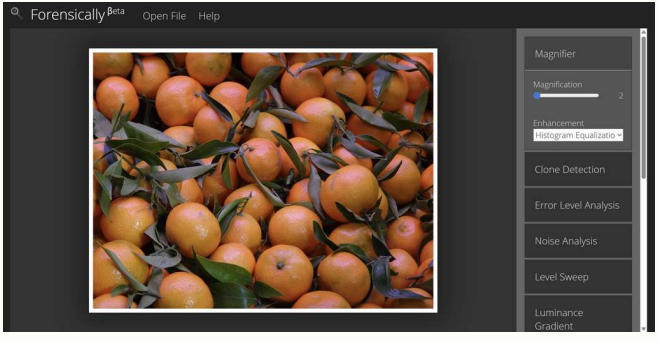
- Width = 1500 = 00 00 05 DC
- Height = 1125 = 00 00 04 65

My Nervous PPT

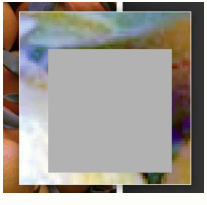
풀이과정



PNG 파일 복원하면 귤 사진이 나옴



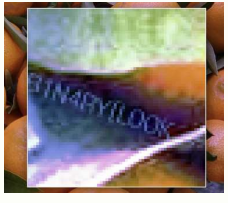


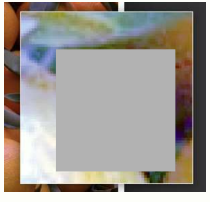


돋보기로 자~세히 살펴보면 문자열이 숨겨져있음

My Nervous PPT

풀이과정





B1N4RY{LOOK [redacted]

최종 Flag

B1N4RY{LOOK[redacted]

사용 도구

HxD Hex Editor

TweakPNG

png-tools

(GitHub: aljazmedic/png-tools)

Forensically

https://dencode.com/ko/string/morse-code